NASONNATION // COMMAND (v7.2)

High-Level System Topology, Distributed AI Compute & Real-Time Streaming Architecture

01 EXECUTIVE ARCHITECTURE OVERVIEW

NasonNation // Command is a self-hosted, zero-cloud-dependency computer vision, facial biometric, and license plate recognition (LPR) platform running on bare-metal Debian Linux (192.168.1.x). Live RTSP camera streams are ingested once by MediaMTX and fanned out simultaneously to web clients over sub-second WebRTC (WHEP) and to an asynchronous Python FastAPI edge CV engine (OpenCV MOG2 + YOLOv8s + YuNet/SFace 128-D embeddings). Heavy multimodal reasoning and plate OCR are offloaded over LAN to a dedicated NVIDIA RTX 3060 Ti node running Qwen2.5-VL via Ollama.

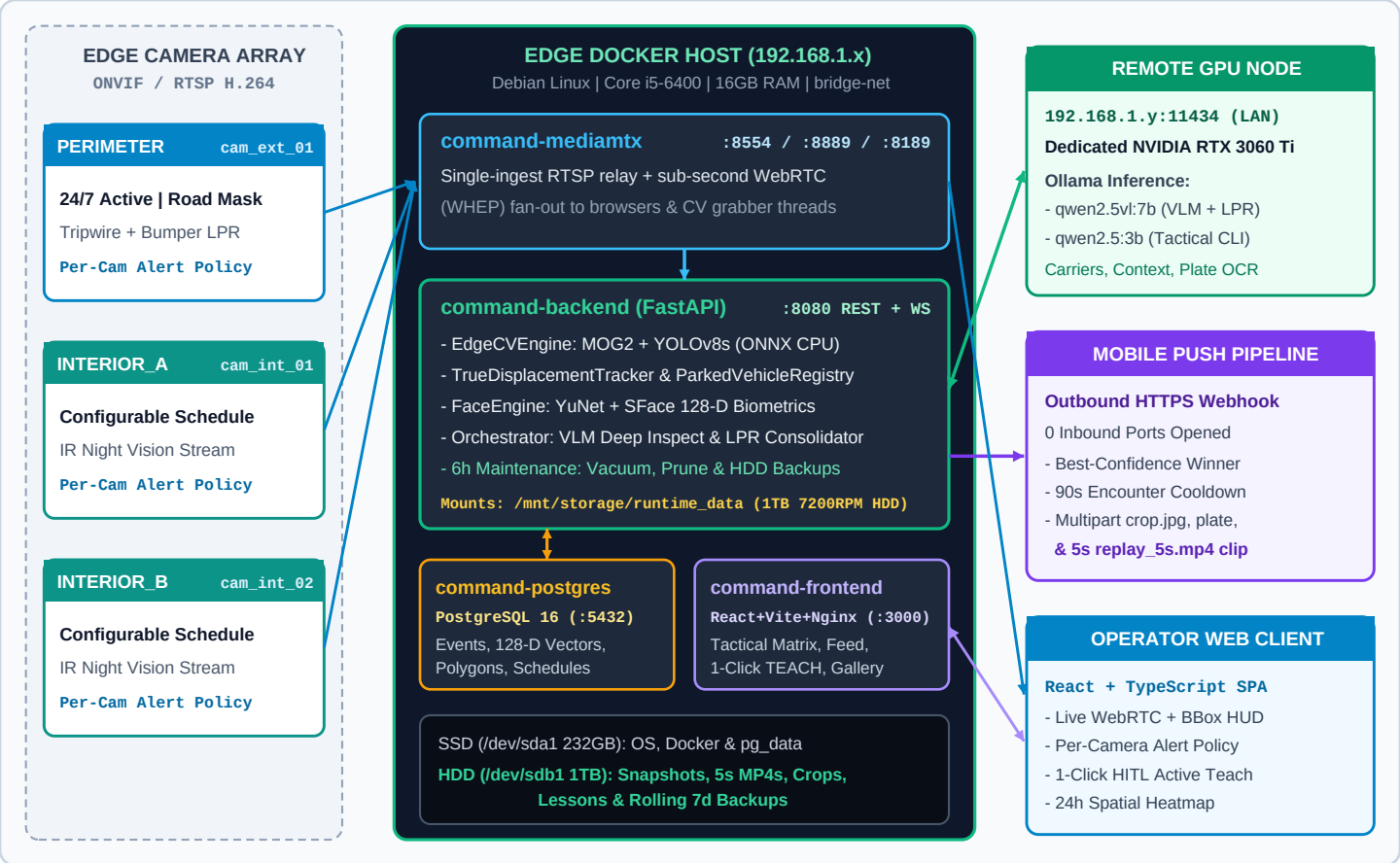
MULTI-STREAM RTSP
24/7 Perimeter + Scheduled Zones

SPLIT-NODE AI
Edge YOLOv8s + RTX 3060 Ti

128-D BIOMETRICS
Cosine Embeddings + RBAC

ZERO OPEN PORTS
Outbound Multipart MP4 Push

02 DISTRIBUTED SYSTEM & NETWORK TOPOLOGY



03 CORE ARCHITECTURAL PRINCIPLES

Split-Compute Efficiency: Lightweight MOG2 motion gating and YOLOv8s ONNX run on the edge CPU, waking the remote RTX 3060 Ti VLM only when verified physical displacement occurs outside road exclusion masks.

Shared Global Biometrics: A single 128-D YuNet + SFace vector gallery in PostgreSQL/RAM serves all camera streams concurrently. Enrolling a trusted user once immediately identifies them across all exterior and interior zones.

Hardware Wear Isolation: High-churn MP4 video clips, high-res snapshots, face crops, and daily compressed backups bind-mount to a 1TB 7200RPM HDD (/dev/sdb1), protecting the 232GB SATA boot SSD (/dev/sda1) from write wear.

04 6-STAGE REAL-TIME DETECTION, BIOMETRIC & VLM PIPELINE

STAGE 1 **RTSP Ring-Buffer & MOG2 Motion Gate**

LatestFrameGrabber maintains a rolling 48-frame (960x540) RAM ring buffer per stream for instant 5-second MP4 pre/post-event replays. OpenCV MOG2 filters static scenes; polygon Exclusion Masks suppress street traffic before neural inference runs.

STAGE 2 **YOLOv8s Detection, True Displacement & Parked-Vehicle Lock**

ONNX Runtime executes YOLOv8s for person, vehicle, and animal classes. ParkedVehicleRegistry locks stationary driveway vehicles after 6 sightings (IoU >= 0.68). Spatial co-occurrence heuristics distinguish walked dogs from solo neighborhood cats.

STAGE 3 **YuNet + SFace 128-D Biometrics & Privacy Guardrails**

Extracts 128-D cosine-similarity embeddings against Trusted Whitelist roles, repeat Visitors (VISITOR_XX), and synced public wanted-notice Watchlist vectors. Role-based privacy guardrails disable external lookup tools on Whitelisted identities.

STAGE 4 **Remote RTX 3060 Ti VLM Inspection & License Plate OCR**

deep_inspect_detection() dispatches target crops + 96%-quality bumper crops (plate_{snap_id}) to Qwen2.5-VL over LAN. Extracts structured JSON summaries, [PLATE: ...] OCR, and Carrier/Service rules (Delivery Alert vs Silent Service).

STAGE 5 **1-Click Human-in-the-Loop Active Learning (TEACH & ID FACE)**

Operator UI provides 1-click corrective preset pills. Confirming a trusted user relabels the event, auto-enrolls 128-D embeddings when facial landmarks are clear, purges false VISITOR_XX entries, and injects camera-specific lessons into VLM prompts.

STAGE 6 **Best-Confidence Encounter Consolidator & Multipart Push**

Awaits burst completion & ffmpeg H.264 encoding (4.5s), evaluates sibling events within +-30s, suppresses bursts matching exempt roles, selects the single highest-confidence frame (90s cooldown), and pushes crop + plate + 5s MP4 via Discord.

05 ROLE-BASED ALERT ROUTING & PER-CAMERA POLICY MATRIX

SUBJECT ROLE / RULE CLASS	MODE: WHITELIST-SILENT	MODE: ALL-ALERTS ACTIVE	MULTIPART PUSH PAYLOAD
PRIMARY_ADMIN (Owner Role)	**SILENT (Breach stripped)**	**SILENT (Admin Exemption)**	Never alerts (0 self-spam)
HOUSEHOLD_WHITELIST (A/B/C)	**SILENT (Trusted Role)**	**ALERTS (Household Arrival)**	Conf % + Crop + 5s MP4
Watchlist Biometric Match	**URGENT ALERT**	**URGENT ALERT**	Priority Ping + Crop + MP4
Perimeter Tripwire Breach	**HIGH ALERT**	**HIGH ALERT**	Priority Ping + Crop + MP4
Logistics Carrier (UPS/AMZN)	**DELIVERY ALERT**	**DELIVERY ALERT**	Carrier + Plate + 5s MP4
Scheduled Service Crew	**SILENT (Visitor crop purged)**	**SILENT (Service Rule)**	Indexed in Event DB only
Unclassified Vehicle / Animal	**Indexed in Event DB only**	**ALERTS (Full-Audit Mode)**	Conf % + Plate + 5s MP4

PER-CAMERA RECORDING & SCHEDULE ENGINE:

Supports 24/7 Continuous, Manual Pause, and Time-Windowed Schedules per stream while preserving live WebRTC viewing.

06 DUAL-DISK STORAGE ARCHITECTURE (NAND SSD WEAR PROTECTION)

TIER 1: /dev/sda1 - 232 GB SATA SSD

Role: Low-Latency OS, Containers & SQL Indexes

- Debian Linux Root (/) & Systemd (fstrim.timer active)
- Docker Engine & Immutable Application Layers (:ro)
- pg_data Volume: PostgreSQL 16 tables & 128-D indexes
- ONNX Runtime & YOLOv8s / YuNet / SFace weights
- JSON Container Logs capped at 10MB x 3 files/service
- Isolated from 24/7 sequential video/image write wear

TIER 2: /dev/sdb1 - 1 TB 7200RPM HDD

Role: Sequential Media, Runtime State & DR Backups

- Dedicated high-endurance sequential storage mount
- runtime_data/snapshots -> /tmp/command_snapshots
  (crop_, full_, plate_ JPEGs + 5s H.264 .mp4 clips)
- runtime_data/faces -> /tmp/command_faces
  (enrolled face crops, delivery_rules & ai_lessons.json)
- runtime_data/backups -> Rolling 7-day .json.gz dumps

07 AUTOMATED SELF-HEALING MAINTENANCE (periodic_maintenance_worker)

1. Bounded Media Quota Enforcement

Audits snapshot storage every 6h; purges clips/crops older than 7 days and enforces a hard 2.0 GB ceiling via FIFO eviction.

2. Event & Ephemeral Vector Pruning

Purges detection_events >14 days old and reclaims 1-sighting transient visitor embeddings unseen for >14 days.

3. Automated VACUUM ANALYZE

Executes autocommit PostgreSQL VACUUM ANALYZE on high-churn tables every 6h to eliminate dead-tuple bloat and maintain query speed.

4. Cross-Drive Disaster Recovery

Writes daily gzipped JSON snapshots (embeddings, polygon zones, schedules, rules, and learned prompts) from Tier-1 SSD to Tier-2 HDD.

08 SERVICE INTERFACES & DEFENSE-IN-DEPTH SECURITY POSTURE

PORT / PROTO	SERVICE / NODE	ARCHITECTURAL PURPOSE & NETWORK BOUNDARY
`3000 / TCP`	**command-frontend**	Private LAN - React/TypeScript Tactical Matrix, Live HUD & Biometric Gallery
`8080 / TCP`	**command-backend**	Private LAN - FastAPI REST (/api/detections, /teach, /faces, /zones) + WS
`8889 / 8189`	**command-mediamtx**	Private LAN - WHEP HTTP Signaling (:8889) & Single-Port WebRTC ICE UDP (:8189)
`8554 / TCP`	**command-mediamtx**	Internal Bridge - RTSP camera ingest & low-overhead OpenCV frame grabbers
`5432 / TCP`	**command-postgres**	Internal Bridge - PostgreSQL 16 (command_db); cross-drive daily backups
`11434 / TCP`	**RTX 3060 Ti Node**	Private LAN - Dedicated Ollama GPU server for Qwen2.5-VL & license plate OCR

SECURITY, OPSEC & GOVERNANCE SUMMARY

1. Zero Inbound Attack Surface: No router port-forwarding; remote mobile alerting operates strictly via outbound TLS webhooks.
2. Secret & Biometric Isolation: Environment secrets, RTSP credentials, database volumes, and biometric crops are excluded from VCS.
3. Immutable Runtime & Privacy Locks: Source mounts read-only (:ro) in containers; external lookup tools auto-lock on trusted roles.